

Standard Authorization Request (SAR)

Complete and submit this form, with attachment(s) to the [NERC Help Desk](#). Upon entering the Captcha, please type in your contact information, and attach the SAR to your ticket. Once submitted, you will receive a confirmation number which you can use to track your request.

The North American Electric Reliability Corporation (NERC) welcomes suggestions to improve the reliability of the bulk power system through improved Reliability Standards.

Requested information			
SAR Title:		Reliability Standards to Address Computational Load - Phase I	
Date Submitted:		February 20, 2026	
SAR Requester			
Name:		Jamie Calderon, Sandhya Madan	
Organization:		NERC	
Telephone:		404-960-0568 470-698-8827	Email: Jamie.Calderon@nerc.net Sandhya.Madan@nerc.net
SAR Type (Check as many as apply)			
☒ New Standard ☒ Revision to Existing Standard ☒ Add, Modify or Retire a Glossary Term ☐ Withdraw/retire an Existing Standard		☐ Imminent Action/ Confidential Issue (SPM Section 10) ☐ Variance development or revision ☐ Other (Please specify)	
Justification for this proposed standard development project (Check all that apply to help NERC prioritize development)			
☐ Regulatory Initiation ☒ Emerging Risk (Reliability Issues Steering Committee) Identified ☐ Reliability Standard Development Plan		☐ NERC Standing Committee Identified ☐ Enhanced Periodic Review Initiated ☐ Industry Stakeholder Identified	
What is the risk to the Bulk Electric System (What Bulk Electric System (BES) reliability benefit does the proposed project provide?):			
This project will address the risks posed to the Bulk-Power System (BPS) due to the emergence of new large loads (e.g., data centers, crypto mining) that are connecting to the grid at an unprecedented scale and speed. As discussed within the 2025 ERO Reliability Risk Priorities Report, these large loads change the landscape of customer demand from organic population-driven growth to rapid investment-based deployment of data centers (including hyperscalers, artificial intelligence (AI), and cryptocurrency), onshoring of manufacturing activities, electrification of industrial processes, and commercial and residential electrification. Some of these loads are asynchronous to the grid, creating system conditions and ride-through challenges during system events. Integrating these large loads and viewing them as BPS resources for flexibility and reliability contribution, when possible, will be challenging but crucial. Other aspects of BPS security and reliability, such as growing uncertainties with global supply chains and development, are similarly impacted. Demand growth from data centers will exacerbate the need for			

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network upgrades and new resources, and uncertainty around supply chains, international trade, import tariffs, and country restriction issues are disruptive to plans and schedules.

To better understand the reliability impact(s) of emerging large loads, NERC's Reliability and Security Technical Committee (RSTC) established the Large Load Task Force (LLTF) and developed a [work plan](#). The Member Representatives Committee provided [written input](#) to this work plan and hosted a [technical panel session](#) at the February 2025 Board meeting. NERC's Board also issued a resolution in February 2025 directing NERC staff to develop an [action plan](#), which will complement the work of the LLTF and provide additional structure to NERC's efforts related to large load integration.

The LLTF released their [Characteristics and Risks of Emerging Large Loads](#) white paper and [summary report of the white paper](#) in July of 2025. This whitepaper similarly identified the impacts that BPS is currently experiencing from large loads and the anticipated increased impacts that large loads will have.

Due to the scale and pace of integrating large loads, NERC is proposing changes to its regulatory framework to have visibility of these impactful BPS users and to enable mitigation of these increasing risks. These changes include: (1) revisions to the Rules of Procedure (ROP) registry criteria, including the development of new NERC registered entity classification consistent with the functional entity classification being developed through registration efforts that include "Computational Loads "; (2) corresponding revisions to the *Glossary of Terms used in NERC Reliability Standards* (NERC Glossary); and (3) the development of new or modified Reliability Standards.

NERC is also developing a Level 3 Alert containing specific actions that NERC has determined are essential for owners, operators, or users of the BPS to take to ensure the reliability of the BPS. NERC expects to take the proposed Level 3 Alert to the NERC Board of Trustees for action in May 2026.

Purpose or Goal (What are the reliability gap(s) or risk(s) to the Bulk Electric System being addressed, and how does this proposed project provide the reliability-related benefit described above?):

The goal of this project will be twofold and require developing: (1) changes to the NERC Glossary to include large load entities, consistent with the proposed modifications for registration criteria in NERC's Rules of Procedure; and (2) a Reliability Standard addressing reliability issues associated with integrating these large loads onto the BPS.

Project Scope (Define the parameters of the proposed project):

The drafting team assigned to this project will coordinate closely with the NERC Staff developing proposed revisions to the NERC Rules of Procedure registry criteria to address the impacts of large loads affecting the reliability of the BPS.

The scope of this project will include:

1. Coordinating with the NERC ROP revisions team to develop new terms and definitions for inclusion in the *NERC Glossary* that align with the proposed changes to the Rules of Procedure registry criteria, to support the development of Reliability Standards applicable to and relating to large loads affecting the reliability of the BPS.
2. Considering the Level 3 Alert Essential Actions (to be issued May 2025), the July 2025 LLTF white paper, and other information and stakeholder feedback, to develop one or more Reliability

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Standard(s) to address the most critical reliability issues associated with integrating these large loads onto the BPS in the near-term. 3. Considering along with NERC staff and the NERC Large Load Working Group, a Phase 2 Standard Authorization Request to address the development of other new or revised Reliability Standards to address computational load needs.
Detailed Description (Describe the proposed deliverable(s) with sufficient detail for a drafting team to execute the project. If you propose a new or substantially revised Reliability Standard or definition, provide: (1) a technical justification¹ of developing a new or revised Reliability Standard or definition, which includes a discussion of the risk and impact to reliability-of the BES, and (2) a technical foundation document (e.g., research paper) to guide development of the Standard or definition): This drafting team will coordinate with the NERC ROP revisions team to develop <i>Glossary</i> definitions corresponding to proposed revisions to the NERC registry criteria under development, including: a. “Computational Load”: <i>tentatively defined as</i> Load comprised of power demand from information technology equipment, such as servers, storage, and networking hardware. b. “Computational Load Entity”: <i>tentatively defined as</i> Entity with Load that includes Computational Load. c. Other defined terms for inclusion in the <i>Glossary</i>, as identified as needed during this project. This drafting team will develop one or more Reliability Standard(s) to address in the near-term essential actions entities must take to assure the reliable integration of large loads into the BPS. This drafting team will also work with NERC Staff and the NERC LLWG to consider if revisions to currently effective or NERC Board of Trustees adopted standards are needed, including existing standards in the MOD, PRC, TOP/IRO, and COM standards families. In the interest of time, these revisions may be pursued as part of a Phase II project. Recommended Approach Due to the urgent nature of the reliability risks to be addressed, NERC Staff recommends the drafting team pursue the development of a new Reliability Standard addressing, in one place, the essential actions that must be addressed in the near term. This approach would allow for a timelier path to mitigation for the most critical issues and present the fewest conflicts with existing and recently completed projects addressing other reliability issues in the Reliability Standards most likely to be implicated (e.g., MOD, TOP/IRO, PRC). Subsequent phase(s) could incorporate and expand upon consensus approaches for addressing large load issues into the existing Reliability Standards, phasing out the initial standard over time as work completes. The drafting team is not bound to Staff’s recommended approach and may revise or develop an alternative approach to addressing the most urgent issues as informed by its expertise, technical information or analysis, or stakeholder feedback.

¹ The NERC Rules of Procedure require a technical justification for new or substantially revised Reliability Standards. Please attach pertinent information to this form before submittal to NERC.

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Cost Impact Assessment, if known (Provide a paragraph describing the potential cost impacts associated with the proposed project):
Cost impacts are unknown currently. NERC Staff anticipates there may be significant cost impacts, although the costs for implementing specific requirements are likely to vary by entity and region.
Please describe any unique characteristics of the BES facilities that may be impacted by this proposed standard development project (e.g., Dispersed Generation Resources):
This project proposes to establish requirements applicable to users of the BPS who are impactful to BPS reliability, and entities who interface with such users.
To assist the NERC Standards Committee in appointing a drafting team with the appropriate members, please indicate to which Functional Entities the proposed standard(s) should apply (e.g., Transmission Operator, Reliability Coordinator, etc. See the NERC Rules of Procedure Appendix 5A:
Members should have experience in operating large load facilities, interfacing with large load facilities, or studying large load facilities in the operations or planning time horizons. Members should also include representatives with experience in compliance or registration matters due to the overlap with the NERC registry criteria revision process.
Do you know of any consensus building activities ² in connection with this SAR? If so, please provide any recommendations or findings resulting from the consensus building activity.
NERC's Reliability and Security Technical Committee established the LLTF to analyze the reliability impacts related to emerging large loads (e.g., data centers (including cryptocurrency mining and AI), industrial facilities, and hydrogen production facilities).
The LLTF was also tasked with defining large loads, and is currently working on a reliability gap analysis white paper to provide more insight on where the most significant risks exist.
Are there any related standards or SARs that should be assessed for impact as a result of this proposed project? If so, which standard(s) or project number(s)?
No.
Are there alternatives (e.g., guidelines, white paper, alerts, etc.) that have been considered or could meet the objectives? If so, please list the alternatives with the benefits of using them.
A large loads white paper was prepared in July 2025. A Level 2 Alert was issued in September 2025 to address reliability concerns related to rapid, major load swings. The alert assessed industry readiness and provided recommendations to help mitigate risks to BPS reliability resulting from integrating Large Loads to the BPS. A Level 3 – Essential Actions Alert is being prepared for NERC Board of Trustees consideration in April 2026.
Given the breadth and extent of the reliability issues, these measures represent an important start but are insufficient to fully address the risks. Full mitigation requires updating the NERC Rules of Procedure registry criteria and developing targeted Reliability Standards to require risk mitigating action.

² Consensus building activities are occasionally conducted by NERC and/or project review teams. They typically are conducted to obtain industry inputs prior to proposing any standard development project to revise, or develop a standard or definition.

Reliability Principles	
Does this proposed standard development project support at least one of the following Reliability Principles ? Please check all those that apply.	
☒	1. Interconnected bulk power systems shall be planned and operated in a coordinated manner to perform reliably under normal and abnormal conditions as defined in the NERC Standards.
☐	2. The frequency and voltage of interconnected bulk power systems shall be controlled within defined limits through the balancing of real and reactive power supply and demand.
☒	3. Information necessary for the planning and operation of interconnected bulk power systems shall be made available to those entities responsible for planning and operating the systems reliably.
☐	4. Plans for emergency operation and system restoration of interconnected bulk power systems shall be developed, coordinated, maintained and implemented.
☐	5. Facilities for communication, monitoring and control shall be provided, used and maintained for the reliability of interconnected bulk power systems.
☐	6. Personnel responsible for planning and operating interconnected bulk power systems shall be trained, qualified, and have the responsibility and authority to implement actions.
☐	7. The security of the interconnected bulk power systems shall be assessed, monitored and maintained on a wide area basis.
☐	8. Bulk power systems shall be protected from malicious physical or cyber attacks.

Market Interface Principles	
Does the proposed standard development project comply with all of the following Market Interface Principles ?	Enter (yes/no)
1. A reliability standard shall not give any market participant an unfair competitive advantage.	Y
2. A reliability standard shall neither mandate nor prohibit any specific market structure.	Y
3. A reliability standard shall not preclude market solutions to achieving compliance with that standard.	Y
4. A reliability standard shall not require the public disclosure of commercially sensitive information. All market participants shall have equal opportunity to access commercially non-sensitive information that is required for compliance with reliability standards.	Y

Identified Existing or Potential Regional or Interconnection Variances	
Region(s)/ Interconnection	Explanation
n/a	n/a

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Risk Tracking	
☒ Grid Transformation ☐ Resilience to Extreme Events ☒ Security	☒ Energy Policy ☐ Critical Infrastructure Interdependencies
CMEP Feedback	
☐ Yes	☐ No

Version History

Version	Date	Owner	Change Tracking
1	June 3, 2013		Revised
1	August 29, 2014	Standards Information Staff	Updated template
2	January 18, 2017	Standards Information Staff	Revised
2	June 28, 2017	Standards Information Staff	Updated template
3	February 22, 2019	Standards Information Staff	Added instructions to submit via Help Desk
4	February 25, 2020	Standards Information Staff	Updated template footer
5	August 14, 2023	Standards Development Staff	Updated template as part of Standards Process Stakeholder Engagement Group
6	June 4, 2024	Standards Information Staff	Updated link to the NERC Reliability Principles
7	July 14, 2025	Standards Information Staff	Minor edits to template
8	July 28, 2025	Standards Information Staff	Updated NERC Only section to match 2025 RISC report.